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Journal of Colloid and Interface Science

Date: December 2025

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Light-trapping superhydrophobic coatings with switchable wettability to solve low-temperature anti-icing/deicing and high-temperature overheating problems on surfaces

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10.1016/j.jcis.2025.138275 ↗

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Highlights

- Light-trapping superhydrophobic coatings (PSTM) prepared were developed.
- PSTM has long-lasting anti-icing/deicing.
- PSTM enables switchable wettability.
- PSTM has low-temperature resistant properties.
- PSTM can solve surface overheating problems.

Abstract

Excessive ice accumulation can cause significant economic losses and even threaten life safety. Therefore, it is crucial to explore integrated strategies for ice prevention and deicing in order to prevent these losses. In this study, a fluorine-free light-trapping superhydrophobic anti-icing/deicing coating with wettability conversion was prepared using modified TiO₂ and multi-walled carbon nanotubes (MWCNTs). The modified TiO₂ nanoparticles and MWCNTs endowed the coating surface with micro/nano rough structures and photothermal properties. The prepared coating has excellent superhydrophobic properties with a contact angle (CA) of 161.5°, which allows the water droplet to delay freezing for about 19 times longer than that of the bare substrate. Furthermore, even at −15 °C, the droplet can still bounce on the coating surface, preventing water accumulation and freezing. Under simulated solar radiation, the frozen water droplet on the surface can be rapidly melted within 283s, thanks to the exceptional multiscale topography and outstanding photothermal performance of the coating surface. Surprisingly, the surface can realize a superhydrophobic to hydrophilic wettability transition after 30 min of UV irradiation, and the superhydrophobicity can be restored after stopping UV irradiation, based on which the idea of controlling the surface temperature by adjusting the wettability of the coating has been proposed, which is expected to solve the problem of overheating of the surface due to high temperatures.

Graphical abstract

4. Conclusion

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgments

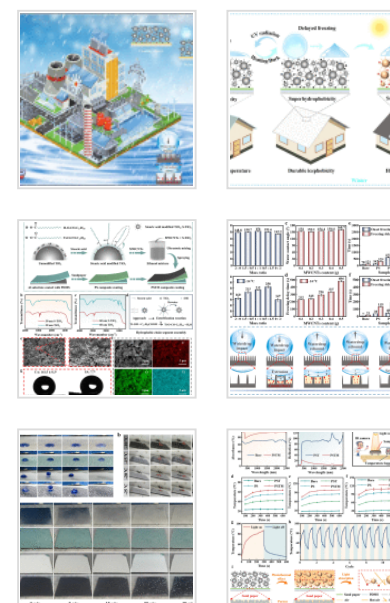
Appendix A. Supplementary data

Data availability

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Keywords

Photothermal; Superhydrophobic; Anti-icing; Deicing; Switchable wettability

Abbreviations

CA, Contact angle; SA, Sliding angle; SLIPS, Slippery liquid-infused porous surface; SHS, Superhydrophobic surface; S-TiO₂, Stearic acid modified TiO₂; Bare, Bare substrate; PS, PDMS pasted sandpaper; PST, PS sprayed with S-TiO₂ solution; PSTM, PS sprayed with mixed solution of S-TiO₂ and MWCNTs

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1. Introduction

Icing is a common physical phenomenon in daily life. Ice typically forms easily on solid surfaces in subzero environments. However, excessive accumulation of ice can cause

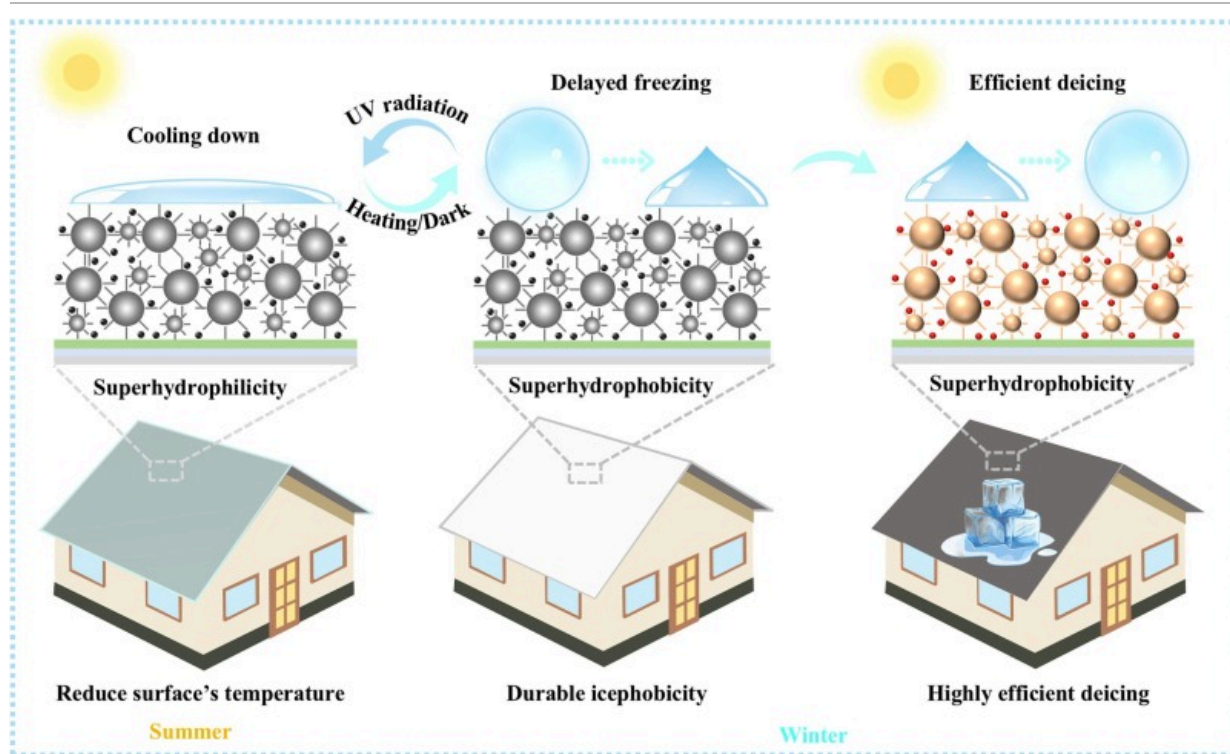
serious economic damage and human injury to society [[1], [2], [3], [4]]. For example, icing on airplane wings affects their dynamics [[5], [6], [7]], icing on turbines reduces power output [[8], [9], [10], [11]], and icing on power lines can lead to circuit blocking [12,13]. Therefore, there is an urgent need to explore efficient anti-icing/deicing strategies to prevent the hazards associated with ice buildup.

Traditional deicing methods include mechanical deicing, electric heating deicing, chemical deicing, etc. These methods require additional technology or energy consumption, may damage material surfaces, are relatively inefficient, or cause environmental pollution. Fortunately, several new methods have been proposed to delay the formation of ice. It has been reported that slippery liquid-infused porous surfaces (SLIPS) not only reduce the accumulation of moisture before freezing but also reduce the ice adhesion strength after freezing, which has attracted significant attention from society [[14], [15], [16], [17], [18]]. However, the anti-icing ability of lubricated surfaces will gradually decline as the lubricant evaporates [19]. Based on this, a superhydrophobic surface (SHS) with a hierarchical texture has been further developed. Water on SHS can only form droplets on the air layer at the solid-liquid interface, which not only reduces the contact area between the droplets and the substrate but also forms a heat conduction barrier to the droplets, making SHS an effective anti-icing material [[20], [21], [22], [23], [24], [25]]. Although SHS has strong ice-delaying properties, ice formation is inevitable when exposed to low temperatures for extended periods of time. Once the rough structure of SHS is wetted, the surface will completely lose its superhydrophobicity and ice resistance, and the water droplets will easily adhere to the surface and freeze quickly. Additionally, the surface microstructure allows water droplets to create a pinning effect, further increasing the adhesion strength between the ice droplets and the substrate [26,27]. Therefore, it is difficult to react to the complex and changing natural environment with only a single passive anti-icing strategy.

As an abundant renewable energy source, solar energy has many advantages in solving globalized energy problems, and it is a feasible way to combine photothermal materials using solar energy as an external input energy source with SHS to form active and passive

combined anti-icing/deicing materials [[28], [29], [30], [31], [32]]. Zhang et al. [33] used double-sized Fe_3O_4 nanoparticles to create a corrosion-resistant photothermal superhydrophobic coating. Under 1 standard sunlight irradiation ($1 \text{ sun} = 1 \text{ kW/m}^2$), the surface temperature could rapidly increase to 89.7°C and the frost layer could be melted within 300s. Photothermal materials can be well used for deicing and defrosting due to their excellent photothermal properties, but this inevitably triggers the problem of overheating of the coating surface in high-temperature environments, which greatly restricts the large-scale application of anti-icing/deicing materials.

In this study, double-sized modified TiO_2 was chosen to create a rough structure, and MWCNTs were used as the photothermal material. Subsequently, a fluorine-free photothermal superhydrophobic coating with switchable wettability was prepared using a simple spraying method, which can be used for long-term anti-icing/deicing at low-temperature environments in winter and for solving surface overheating problems caused by high temperatures in summer (Fig. 1). As a result, the coating exhibited a contact angle (CA) of 161.5° and a sliding angle (SA) of 7.7° . At -10°C , the coating showed good resistance to ice, with an average onset freezing time of 1756s and an average freezing delay time of 2248s. MWCNTs improve the light absorption performance and light-heat conversion ability of the coating, and the superhydrophobic rough microstructure allows sunlight to be captured by multiple internal reflections, resulting in highly efficient deicing of 283s. Additionally, through UV irradiation, it was verified that the coating had a wettability conversion function, allowing it to switch freely between hydrophobicity and hydrophilicity, and the hydrophilic state of the coating demonstrated an efficient cooling effect of 53.8°C . This provides a viable, economic, and environmentally friendly solution to avoid overheating of coatings due to high temperatures in hot summers, greatly broadening the application range of anti-icing/deicing coatings.



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Fig. 1. Function demonstration diagram of PSTM.

2. Experimental details

2.1. Chemicals and materials

In this study, the chemicals and materials required were detailed in the Supporting Information note S1 (Chemicals and materials).

2.2. Preparation of sample

2.2.1. Hydrophobic modification of TiO₂ nanoparticles

To perform hydrophobic modification of titanium dioxide nanoparticles (TiO₂), 1 g of stearic acid was added to a mixture of deionized water (25g), ethanol (40g) and TiO₂ (2g) nanoparticles, followed by magnetic stirring at 30 °C for 3h. After stirring, we dried the mixture at 125 °C for 3h to obtain hydrophobically modified TiO₂ (S-TiO₂).

2.2.2. Preparation of light-trapping superhydrophobic coatings

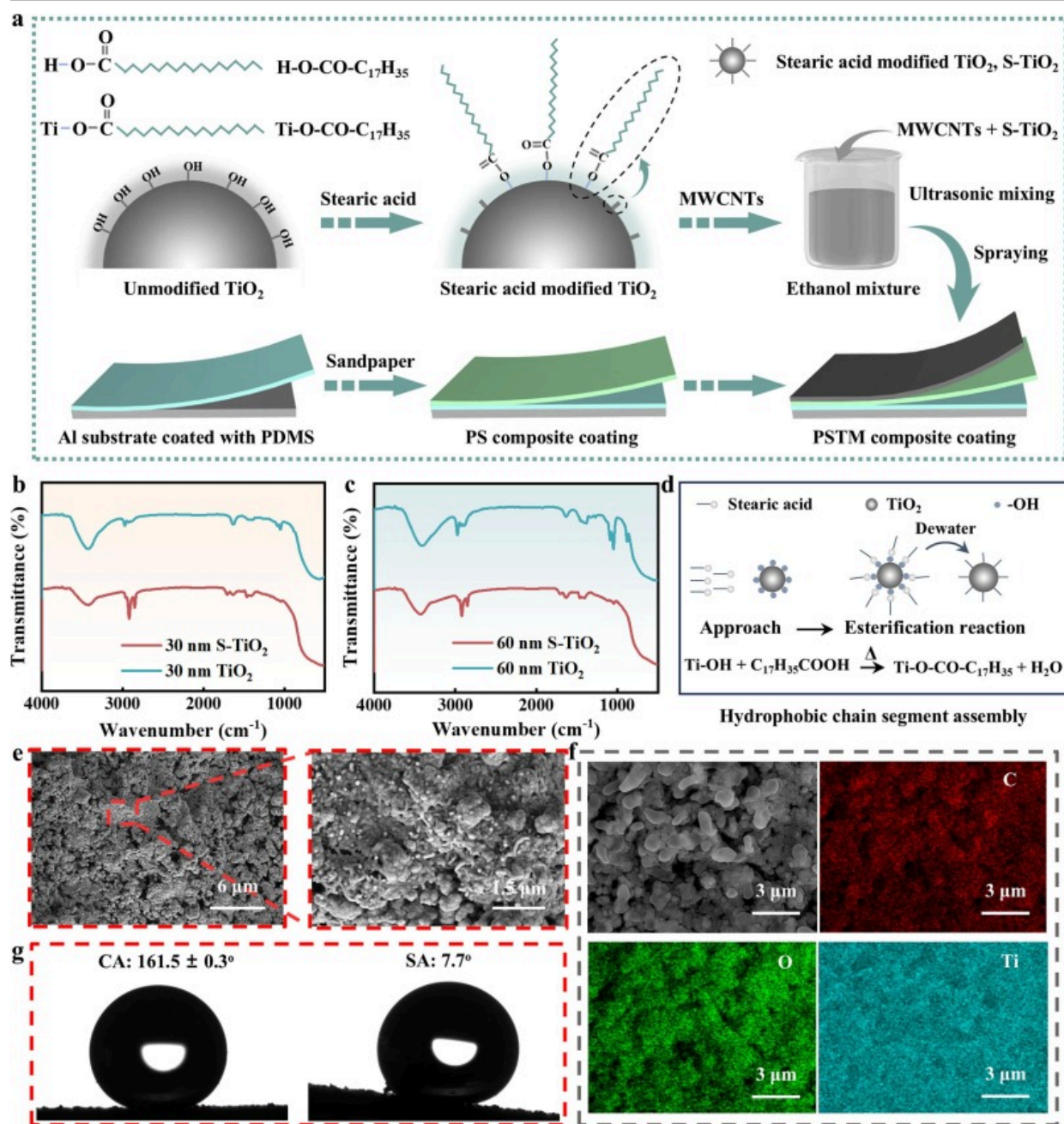
Firstly, we dispersed polydimethylsiloxane (PDMS) and curing agent in ethyl acetate at a mass ratio of 10 : 1 and mechanically stirred it with a glass rod for 5min until the PDMS was completely dissolved, and then the sandpaper was adhered to an aluminum plate (Bare) with PDMS and cured for 40min at 80 °C to form the thermal insulation layer (PS). Next, we mixed 2g of S-TiO₂ (the mass ratio of 30nm and 60nmS-TiO₂ was 1 : 1) with 14g of ethanol and sonicated for 15min to obtain the S-TiO₂ mixture; furthermore, we added multi-walled carbon nanotubes (MWCNTs) to the S-TiO₂ mixture and continued to sonicate for 15min to obtain a mixture of MWCNTs and S-TiO₂, which was sprayed by using a gravity. The S-TiO₂ mixture and the mixture of MWCNTs and S-TiO₂ were uniformly sprayed on the PS surface using a gravity spray gun (spraying air pressure of 2.4bar, nozzle type of atomizing nozzle), and cured at 80 °C for 40min to obtain the superhydrophobic coating (PST) and photothermal superhydrophobic coating (PSTM).

3. Results and discussion

3.1. Microscopic morphology and wettability analysis

In this study, a superhydrophobic anti-icing/deicing coating with photothermal properties was successfully fabricated by hydrophobic modification of TiO₂ nanoparticles and combination with MWCNTs using a simple spraying process (Fig. 2a). Figs. 2b and c present images of the Fourier transform infrared spectroscopy (FTIR) of 30nm and 60nm TiO₂ before and after modification. In Fig. 2b, there is an obvious absorption peak at

3414cm^{-1} in both curves (tensile vibration of —OH), and the peak of hydroxyl absorption of S-TiO_2 is clearly weakened, which indicates a reduction in the number of hydroxyl groups on the surface of S-TiO_2 , suggesting that the esterification of TiO_2 with stearic acid may have occurred. S-TiO_2 shows new absorption peaks at 2918cm^{-1} and 2850cm^{-1} , which could be attributed to the C—H bond in —CH_3 and —CH_2 of stearic acid. Meanwhile, the peak located at 1708cm^{-1} is ascribed to the C=O bond in —COOH . These results indicate the chemical bonding of TiO_2 (30nm) with stearic acid. Fig. 2c shows the FTIR of 60nm TiO_2 and S-TiO_2 , which is consistent with the trend of change in Fig. 2b, indicating the successful grafting of low surface energy groups on the 60nm TiO_2 as well. Further analyzing the mechanism of hydrophobic modification of the TiO_2 surface, as shown in Fig. 2d, the stearic acid molecule approaches the —OH on the TiO_2 surface through the carboxylic acid head, and under the condition of heating, the —COOH of the carboxylic acid and the —OH of TiO_2 are dehydrated and condensed, and the ester bond (TiO—O—CO—) is generated. At this time, the silane hydrophobic chain segments of stearic acid are attached to the TiO_2 surface through ester bonds, and the outwardly oriented arrangement of the alkyl chains imparts hydrophobic properties to S-TiO_2 .



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Fig. 2. Preparation and characterization of PSTM coatings. (a) The preparation progress of PSTM coating. FTIR spectra of (b) 30nm and (c) 60nm TiO₂ before and after modification. (d) Mechanism of hydrophobic modification of TiO₂. (e) SEM images and (f) EDS mapping images of PSTM. (g) The CA and SA images of PSTM.

According to the Cassie-Baxter model, low surface energy and multiscale surface roughness are the two primary factors for superhydrophobic surfaces. The scanning electron microscopy (SEM) image (Fig. 2e) shows that the surface layer of PSTM is uniformly covered with a large number of dense nanoparticles. Combined with the energy dispersive spectroscopy (EDS) image analysis (Fig. 2f), these nanoparticles are mainly S-TiO₂ and MWCNTs. It can be seen from Fig. 2e that there are many double-sized S-TiO₂ nanoparticles agglomerated into large micron-sized particle clusters, forming a structure similar to a 'soil' pile-up, while MWCNTs are 'tree roots' buried deep in the 'soil', which serve as good connectors between the S-TiO₂. This micro/nano multiscale morphological property gives the surface a rough microstructure, and in combination with the alkyl hydrophobic chain segments on the surface of S-TiO₂ particles, a superhydrophobic micro-textured surface was successfully constructed, resulting in excellent superhydrophobicity with a high contact angle (CA) of 161.5° and a low sliding angle (SA) of 7.7° (Fig. 2g). Since PSTM exhibits superior non-infiltrating superhydrophobicity, it is expected to have extensive applications in passive anti-icing.

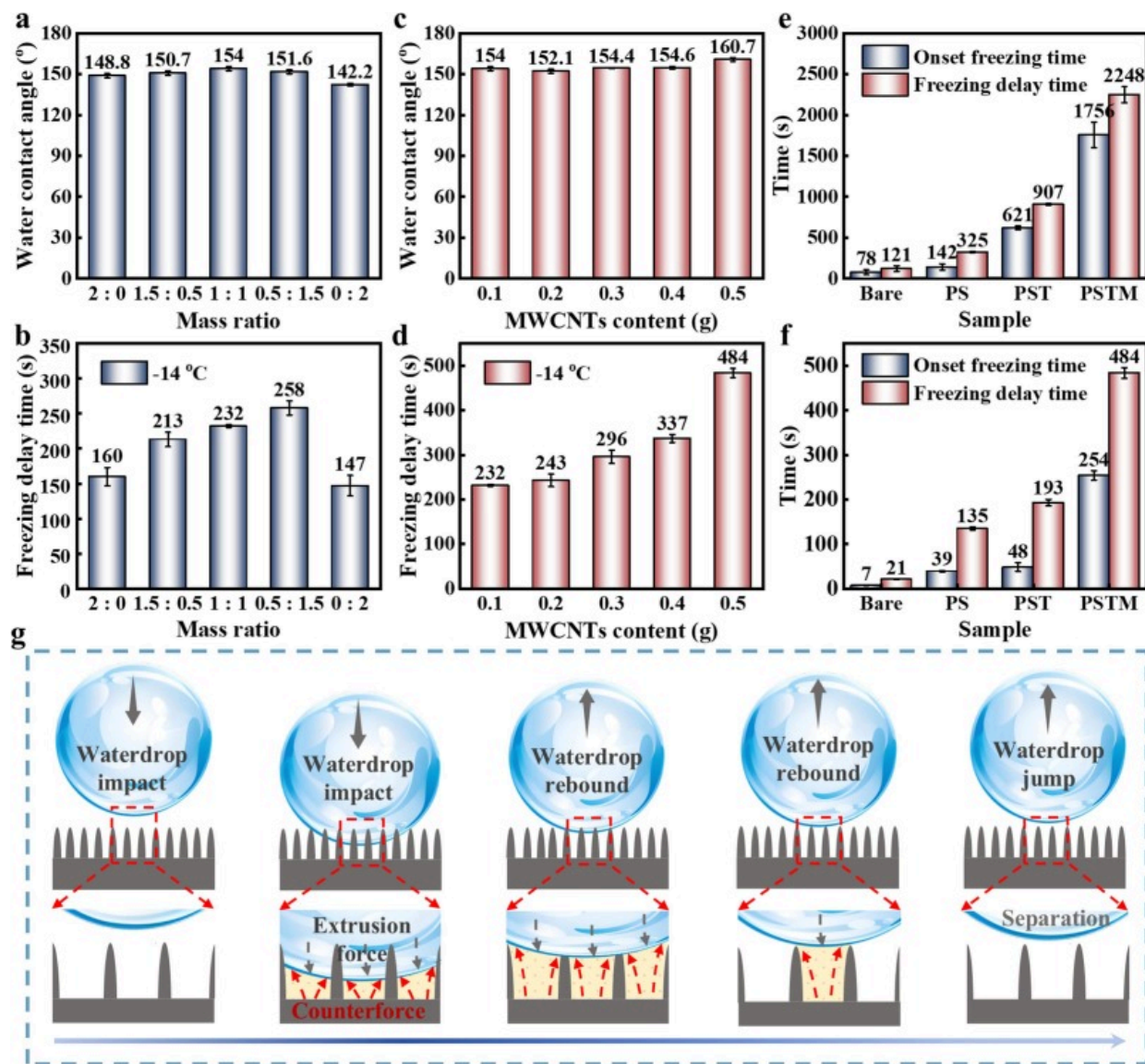
3.2. Anti-icing and anti-frosting performance at low temperatures

3.2.1. Anti-icing performance

As shown in Fig. S1, droplet freezing is a dynamic process in which the droplet transitions from an initially completely transparent state (state 1) to a semi-transparent state (state 2), and then from the semi-transparent state to a completely opaque frozen state (state 3), at which point an ice spike appears at the top of the ice droplet. Based on observations of the aforementioned freezing process, to quantitatively assess the anti-icing performance of materials, we developed an anti-icing/deicing platform (Fig. S2) and proposed a

quantitative analysis method based on freezing time. Specifically, the time required for the water droplet to transition from state 1 to state 2 is defined as the onset freezing time, while the time required for the water droplet to transition from state 1 to state 3 (i.e., the final ice formation state) is defined as the freezing delay time, also known as the total freezing time. During the coating design process, it was found that the content of S-TiO₂ and MWCNTs influences the hydrophobicity of the PSTM surface by affecting the construction of the surface microstructure, which ultimately impacts its anti-icing performance. Given this relationship, this study employed a controlled variable method to separately investigate the effects of S-TiO₂ and MWCNTs on the hydrophobicity and anti-icing performance of the PSTM surface. Firstly, the experiment fixed the MWCNTs content at 0.1 g and adjusted the mass ratio of the dual-scale S-TiO₂ (30 nm : 60 nm) to investigate the effect of the dual-scale S-TiO₂ content on the surface hydrophobic angle and anti-icing performance of the PSTM. As shown in Fig. 3a, when the mass ratio of the two sizes of S-TiO₂ is 1 : 1, it is more conducive to the formation of a superhydrophobic surface (CA: 154°). This may be because the synergistic effect of the 1 : 1 mass ratio of dual-scale S-TiO₂ more easily forms a multi-level rough structure, where larger nanoparticles form the primary rough structure, and smaller nanoparticles fill the gaps to form the secondary roughness. This multi-level rough structure significantly increases the surface area and air retention capacity of the PSTM, effectively raising the energy barrier between the solid and liquid phases, hindering heat exchange between droplets and the surface, thereby delaying the freezing process and increasing freezing time (Fig. 3b). Based on the above results (optimal hydrophobicity and delayed icing performance), the mass ratio of S-TiO₂ was determined to be 1 : 1 as the optimal ratio for subsequent studies. Subsequently, under the condition of a fixed S-TiO₂ mass ratio of 1 : 1, the effect of MWCNTs content (0.1 g to 0.5 g) on the hydrophobicity and delayed icing performance of PSTM was investigated. An interesting phenomenon was observed, as shown in Fig. 3c: when the mass of MWCNTs was increased in a non-linear manner from 0.1 g to 0.5 g, the CA of droplets on the PSTM surface increased non-linearly. This indicates that within this addition range, MWCNTs effectively increase the surface roughness of PSTM, resulting in a positive correlation between the CA and MWCNTs addition. According to classical

nucleation theory, the larger the CA on a solid surface, the higher the free energy barrier for ice crystal nucleation, the lower the nucleation rate, and the more significant the delayed icing effect on the surface. Therefore, it is reasonable that the longest freezing delay time was observed when the MWCNTs addition was 0.5 g (Fig. 3d). Considering hydrophobicity, anti-icing performance, material solubility, and the feasibility of the spraying process, this study ultimately selected 0.5 g MWCNTs and 2 g S-TiO₂ (with a mass ratio of 30 nm : 60 nm S-TiO₂ of 1 : 1) as the optimal mass parameters for preparing PSTM.



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Fig. 3. Hydrophobicity and anti-icing properties. The effect of the mass ratio of S-TiO₂ on (a) the CA and (b) anti-icing performance of the PSTM surface. The effect of MWCNTs content on (c) the CA and (d) anti-icing performance of the PSTM surface. Freezing time

of water droplets on Bare, PS, PST, and PSTM sample surfaces at (e) -10°C and (f) -14°C . (g) Dynamic process of water droplets impacting the PSTM surface.

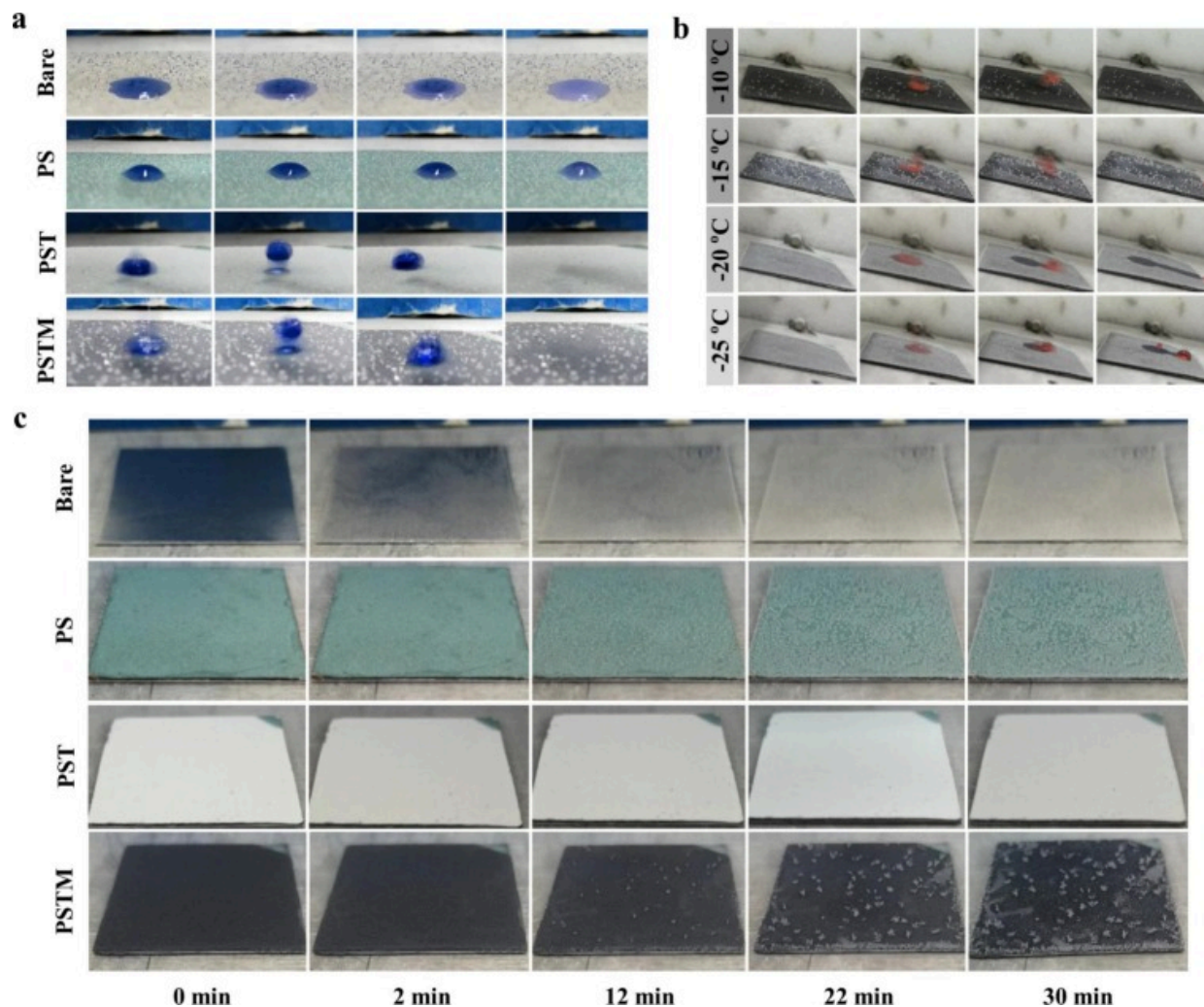
Furthermore, to evaluate the anti-icing performance of the optimized PSTM, we observed the icing process of droplets on its surface and those of different control samples and measured the icing time of the droplets at -10°C and -14°C . As shown in Figs. S3 and 3e, at -10°C , the initial temperature of the bare substrate surface and the droplets was significantly lower than that of other samples, and its total freezing time was the shortest (121 s). In contrast, the onset freezing time of droplets on the PSTM surface was the longest, reaching 1756 s, approximately 23 times that of the bare substrate; its freezing delay time was also the longest, at 2248 s, approximately 19 times that of the bare substrate. At -14°C , the initial temperature of the bare substrate surface and droplets remained the lowest, and the total freezing time of the droplets remained the shortest (21 s), while the total freezing times for PS, PST, and PSTM were 135 s, 193 s, and 484 s, respectively (Figs. S1, S4, and 3f). Notably, the anti-icing performance of PSTM also stands out among recently reported anti-icing surfaces (Fig. S5). The excellent anti-icing performance of PSTM is primarily attributed to its multi-scale rough superhydrophobic structure. This structure traps a large amount of air between the PSTM surface and the liquid, forming an air layer (air pocket), which not only significantly increases thermal conductivity resistance but also reduces the actual contact area between the liquid and the solid surface, thereby greatly reducing the thermal conductivity efficiency between the water droplet and the interface. In stark contrast, the aluminum substrate has excellent thermal conductivity ($237\text{ W}/(\text{m}\cdot\text{K})$) and a smooth, flat surface, enabling heat to be rapidly transferred from the cooling plate to the water droplets through the aluminum substrate, thereby accelerating their solidification. The structural design of PSTM further enhances its thermal insulation performance: the photothermal hydrophobic layer is separated from the aluminum substrate by PDMS and sandpaper, effectively inhibiting the heat transfer path from the cooling plate to the water droplets. Therefore, the PSTM surface can effectively suppress ice nucleation, hinder ice crystal growth, and exhibit significant anti-icing capabilities; even after the droplet is completely frozen, it can

maintain an approximate spherical shape, which directly reflects its low surface adhesion characteristics. In addition to static anti-icing performance, we also studied the dynamic anti-icing performance of the PSTM surface at -10°C . After 30min of continuous droplet impact (Video S1), the surface remained ice-free. This is primarily due to the surface's microscopic rough structure: when droplets impact the surface, the air layer within the structure exerts an upward elastic repulsive force on the droplets, causing them to quickly bounce off the surface (Fig. 3g); additionally, the droplets' own gravitational force also aids in their escape from the surface.

3.2.2. Low-temperature resistance

To further validate the low-temperature tolerance of the samples, we placed them in a low-temperature environment for 30min and observed the bouncing behavior of liquid droplets on their surfaces (see Supporting Information note S2.5 (Low-temperature resistance test) for specific experimental details). As shown in Fig. 4a, at -10°C , liquid droplets on the surfaces of PST and PSTM could detach from the surfaces through bouncing behavior, indicating that both materials could maintain their hydrophobicity for at least 30min at this temperature, demonstrating excellent low-temperature anti-icing potential. However, PST fails to meet the requirements for active deicing due to its insufficient photothermal performance. In contrast, PSTM not only inherits the excellent anti-icing properties but also possesses enhanced photothermal performance, enabling it to achieve efficient active deicing and thus meet the comprehensive anti-icing and deicing application requirements under harsh environmental conditions. Furthermore, to systematically study the effect of temperature on the dynamic behavior of droplets on the PSTM surface (particularly their detachment capability under gravitational driving), we applied a 15° inclination angle to the sample surface (imparting an initial velocity for droplets to slide along the surface) and observed their behavior after being placed in different low-temperature environments (-10°C to -25°C) for 30min. Fig. S6 records the initial temperatures of the PSTM surface under different low-temperature conditions. As the environmental temperature decreases, the initial temperature of the sample surface decreases, and the bouncing behavior of droplets on the PSTM surface gradually weakens.

As shown in Fig. 4b, at -10°C and -15°C , droplets can quickly leave the surface after contacting the PSTM surface and undergoing multiple bounces, with no residual droplet particles, indicating that the surface maintains good hydrophobicity; at -20°C , the droplets do not exhibit obvious bouncing but can still leave the surface by sliding; surprisingly, even at -25°C , the droplets can still slide a certain distance. Based on the experimental results, PSTM demonstrates excellent low-temperature tolerance: even under low-temperature icing conditions, its surface can maintain the droplets' ability for directed movement (bouncing or sliding), enabling the droplets to detach from the surface. The core mechanism behind this phenomenon lies in the air pockets formed by the Cassie-Baxter state maintained between water droplets and uneven surface structures, combined with gravitational forces, which collectively promote the dynamic detachment behavior of droplets. This sustained directed movement effectively prevents water droplets from accumulating and freezing on the surface, significantly enhancing the dynamic anti-icing performance of PSTM.



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Fig. 4. Low-temperature resistance and anti-frosting performance. (a) Bouncing behavior of water droplets on different sample surfaces at -14°C . (b) Bouncing behavior of water droplets on the PSTM surface at different temperatures. (c) Frosting processes on different sample surfaces at -15°C .

3.2.3. Anti-frosting performance

Considering practical applications, the sample was placed on the cooling plate of a -15°C refrigeration equipment (with an ambient fog temperature of -12.8°C) to observe frost formation on the sample surface. Fig. S7 records the initial temperature of the samples, and Fig. 4c shows the frosting process on the surface in a dry state (without added water droplets). It can be observed that after 30min, frost layers formed on all surfaces, but the frost layer thickness on the PSTM surface was significantly lower than that on other samples. Due to the low energy barrier for nucleation on the bare substrate surface, the frost layer grows and expands after the initial frost embryo on the surface, and the entire surface is almost completely covered with frost within 2min, and a dense frost layer forms on the bare substrate surface after 30min. In contrast, on a superhydrophobic surface, a higher energy barrier must be overcome to initiate nucleation [34,35]. The rough surface structure and low thermal conductivity of the PSTM surface increase the nucleation barrier for ice crystals, resulting in only a few sites being initiated during the initial heterogeneous nucleation process, which causes a delayed frosting phenomenon on the surface of PSTM. After 30min, although the edges of the PSTM were covered with a layer of frost, the central area remained exposed, confirming its excellent anti-frosting performance. It was also observed that the frost started to form from the edges of PSTM, a result similar to the study of Zhang et al. [36], which can be attributed to the edge effect of frosting. This effect may be caused by a defect at the edge of the superhydrophobic surface, which is caused by the surface cutting or the limitation of the surface treatment process, resulting in surfaces exhibiting hydrophilic properties at the edges, which further leads to preferential frosting at the edges [37,38]. Quantitative data further confirms the anti-frosting performance of PSTM: as shown in Fig. S8, the frost layer accumulation rate per unit area of PSTM is slow, with only $2.85\text{mg}\cdot\text{cm}^{-2}$ of frost layer accumulated after 40min of exposure at -15°C , highlighting its practical application potential.

3.3. Photothermal and photothermal deicing performance

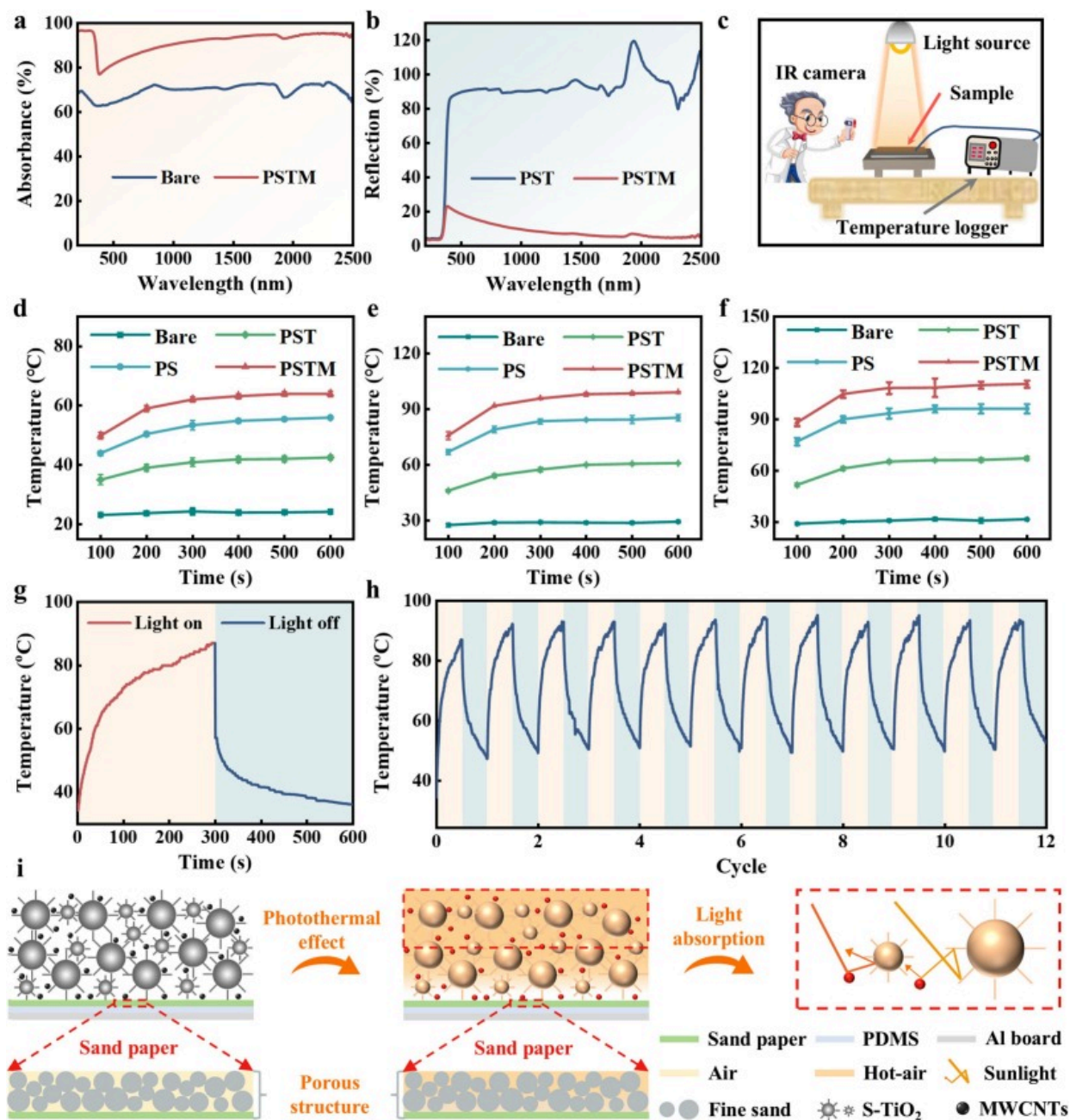
3.3.1. Photothermal performance

Although PSTM superhydrophobic surfaces can show excellent resistance to icing and frosting, building surfaces and outdoor equipment surfaces are frequently susceptible to being heavily covered with a thick layer of ice or frost in low temperatures and high humidity, meaning that ice formation is inevitable. Therefore, deicing and defrosting are vital for the practical implementation of anti-icing/deicing materials. As for photothermal deicing materials, excellent photothermal performance is the key to meeting the actual deicing needs, so this study comprehensively evaluates the photothermal performance of the coatings in terms of both light absorption performance and photothermal conversion performance. The light reflectance (R) of the coating surfaces was tested using a UV–Vis–NIR spectrometer for light wavelengths ranging from 200 to 2500nm, and the light absorption (A) was calculated as follows:

$$A(\%) = 100\% - R(\%) - T(\%) \quad (1)$$

In this study, the substrates of all the samples are aluminum plates, which can be considered opaque, so the transmittance (T) is negligible. Fig. 5a shows that in the wavelength range of 200–2500nm, the average light absorption of PSTM can reach more than 90%, while the average light absorption of Al is only 70%, which is mainly ascribed to the micro/nano rough structure on the surface of the PSTM coating and the excellent light-absorbing ability of MWCNTs. Meanwhile, the higher light reflection of PST is due to the high reflectivity of S-TiO₂ (Fig. 5b). Furthermore, on the basis of Section 3.2.1, the effects of MWCNTs and S-TiO₂ on the photothermal performance of PSTM were also experimentally explored. The experiments verified that the double-scale S-TiO₂ was also favorable for the construction of efficient light-absorbing structures of PSTM (Fig. S9), which may be attributed to the fact that the multilevel rough texture built by the double size can effectively lengthen the traveling path of the light in the structure, thus realizing the multiple reflections of the light, and in combination with the highly efficient photon absorption property of MWCNTs, the surface of the PSTM can sufficiently absorb the photon through multiple reflections and convert it into the heat release. Meanwhile, the

photothermal performance of PSTM showed a significant growth trend with the increase of MWCNTs content (Fig. S10).



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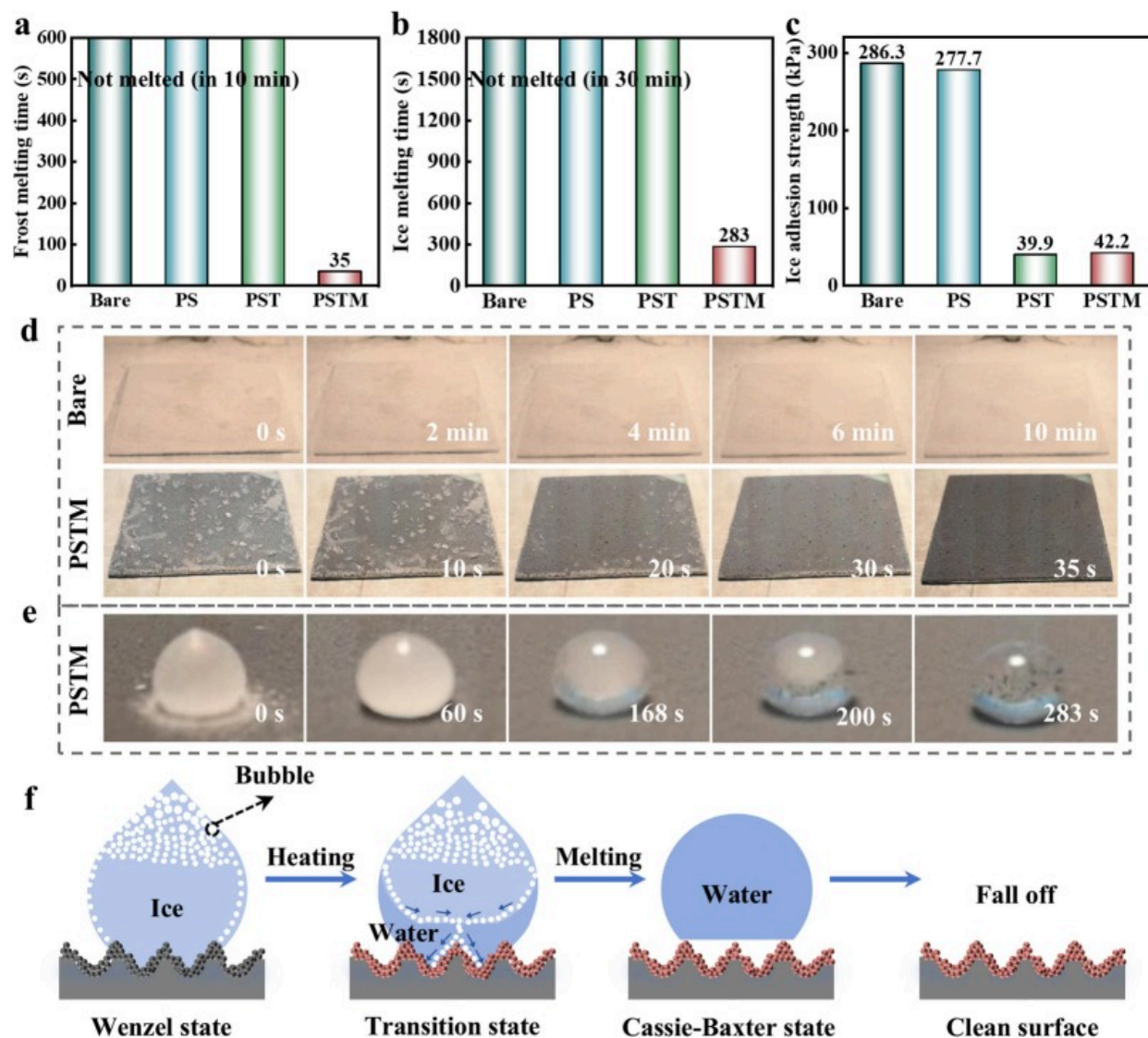
Fig. 5. Photothermal performance. (a) Absorption spectra of Bare and PSTM. (b) Reflection spectra of PST and PSTM. (c) Schematic diagram of the simulated photothermal experiment. (d-f) Surface temperature change curves of bare, PS, PST, and PSTM under 0.5 sun, 1 sun, and 1.5 sun conditions over 600s. (g) Temperature change of PSTM in a single intermittent light exposure test (1 sun). (h) Temperature change of PSTM in 12 intermittent light exposure tests (1 sun). (i) Mechanistic diagram of the photothermal effect of PSTM.

Considering the light intensity is constantly changing with weather conditions in practical applications, the temperature variations of samples' surfaces under different simulated solar radiation (0.5 sun, 1 sun and 1.5 sun) were explored to evaluate the photothermal conversion performance. In the experiment, the samples were exposed to simulated sunlight, and their surface temperatures were monitored in real time using a thermal camera and a temperature data logger (Fig. 5c). As shown in the data from Figs. 5d-f, PSTM consistently maintained the highest temperature under different light intensities (e.g., 99.1 °C after 600s of exposure to 1 sun intensity) and continued to rise as light intensity increased. Its temperature increase significantly surpassed that of traditional photothermal superhydrophobic materials (Fig. S11), confirming the material's exceptional photon capture and photothermal conversion capabilities. Given the importance of photothermal stability for practical applications, further intermittent light exposure cycle tests were conducted on PSTM under 1 sun (each cycle consisting of 300s of heating/300s of cooling, Fig. 3g). The results showed that PSTM's temperature response remained stable over 12 consecutive cycles (Fig. 3h), demonstrating its reliable thermal cycle stability. Additionally, PSTM maintains stable photothermal performance under 0.5 sun and 1.5 sun light intensities (Figs. S12 and S13), providing a solid foundation for its anti-icing/deicing applications in real-world, variable environments. Further analyzing the photothermal mechanism of PSTM, as shown in Fig. 5i, the concave structures on the coating surface increase the specific surface area, and when simulated sunlight irradiates the coating surface, the rough surface structure captures the light by multiple reflections, with the excellent photothermal performance of the MWCNTs together, which not only

improves the light-absorption ability but also endows the coating with an excellent photothermal conversion ability, and thus the PSTM exhibits outstanding photothermal performance. The outstanding photothermal performance of PSTM provides a strong guarantee for the next applications in deicing and defrosting.

3.3.2. Photothermal deicing performance

Photothermal deicing and defrosting performances were tested on samples' surfaces and specific details can be found in the Supporting Information note S2.8 (Photothermal deicing test). Figs. 6a and d demonstrate the frost-melting performance of the samples. The excellent photothermal performance leads to the complete melting of the frost layer on the surface of PSTM after 35 s under 1 sun; however, the frost layers on the surfaces of Bare, PS, and PST remain unmelted after 10 min of light illumination. Furthermore, Figs. 6b and e show that at -14°C , PSTM was able to rapidly melt the surface ice in only 283 s at 1 sun, while the ice on the surfaces of Bare, PS, and PST could not be melted in 30 min. This is mainly attributable to the excellent photothermal performance of the PSTM surface. When light hits the surface, the photothermal effect increases the temperature of the PSTM surface, and the heat is transferred to the ice droplets from the bottom up, causing the ice at the interface between the ice droplets and the solid surface to melt first, forming a film of water until it is all melted, and due to the special superhydrophobic structure, the melted ice can easily slide off from the surface under a certain external force (Fig. 6f). And even without external force, gravity in an inclined position causes the melted ice to roll off the surface, resulting in rapid deicing. All of the above analysis shows that deicing and defrosting become more efficient and energy-saving with the combined synergy of the photothermal effect and superhydrophobicity. In addition, the ice adhesion strength of the samples was tested at -14°C , where the ice adhesion strength of PSTM was 42.2 kPa, which was much lower than that of Bare and PS, further demonstrating superior deicing performance of PSTM (Fig. 6c).



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Fig. 6. Deicing performance of PSTM. (a) Frost melting times of Bare, PS, PST and PSTM at -15°C . (b) Ice melting times of Bare, PS, PST and PSTM at -14°C . (c) Ice adhesion strength at -14°C for Bare, PS, PST, and PSTM, respectively. (d) Melting frost processes of Bare and

PSTM. (e) Melting ice process of PSTM. (f) Schematic representation of the ice melting mechanism on the PSTM superhydrophobic surface.

3.4. Switchable wettability

By introducing photothermal materials, solar energy can be converted into heat energy, effectively solving the issue of deicing and defrosting in winter; however, in hot summer applications, the excellent photothermal properties of the coating may create a new problem: high temperatures leading to surface overheating. In this study, it was observed that the PSTM coating could realize the transition from superhydrophobic to hydrophilic under UV irradiation, and the coating gradually regained the superhydrophobicity after stopping the UV irradiation. Based on this phenomenon, a solution to prevent high-temperature overheating of a coating is proposed herein, i.e., to further regulate the surface temperature of the coating by controlling the wettability transition of the coating from hydrophobic to hydrophilic. This allows for the dual use of the coating in both winter and summer (Figs. S14 and S15), greatly expanding the application range of the coating. Next, the switchable wettability of the PSTM and the cooling performance of the hydrophilic surface will be verified.

As shown in Fig. 7a, the CA on the PSTM surface continuously decreases with increasing UV irradiation time, dropping to approximately 45.1° after 30min, indicating that the coating surface has become hydrophilic. To characterize the rate of surface hydrophilization, we monitored the change in CA rate (η) with irradiation time (calculated at 5-min intervals; detailed information is provided in Supporting Information S2.9). As shown in Fig. S16, the initial η value for the first 5 min was $3.4^\circ/\text{min}$, which then decreased to $1.6^\circ/\text{min}$ during the second 5-min interval; thereafter, η values continued to rise, reaching a peak of $6.4^\circ/\text{min}$ during the final 5 min, indicating the fastest hydrophilization rate at this point. After ceasing UV irradiation, the coating gradually recovered from hydrophilic to superhydrophobic. Based on this, we define the transition of the surface from superhydrophobic to hydrophilic and back to superhydrophobic as a complete wetting conversion test. Fig. 7b shows the change in surface free energy during

a wetting conversion test: from the initial value 0.94 mN/m to the intermediate value 42.94 mN/m, and finally decreasing to 9.86 mN/m, clearly demonstrating the trend of surface energy first increasing and then decreasing. This indicates that the surface properties of PSTM undergo a transition from non-polar dominance to polar dominance and back to non-polar dominance, confirming its successful achievement of hydrophobic-hydrophilic-hydrophobic wettability switching. As shown in Fig. 7c, even after 10 cycles of testing, the surface remains superhydrophobic, verifying the stability of PSTM's wettability switching performance. Additionally, after UV, high-humidity, and thermal cycling tests (Fig. S17), PSTM still exhibits stable wettability switching performance (Fig. 7d), indicating that the S-TiO₂ photocatalytic activity can withstand complex environments. In summary, PSTM achieves a reversible transition from superhydrophobic to hydrophilic upon UV irradiation.



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Fig. 7. Switchable wettability and hydrophilic cooling performance of PSTM. (a) Relationship between the CA of the PSTM surface and UV irradiation time within 30min. (b) Changes in surface energy of PSTM during the wetting cycle. (c) Relationship between CA and the number of wetting cycles. (d) Wettability switching performance of the PSTM surface after high humidity, thermal cycling, and UV aging tests. (e) Schematic diagram of the photocatalytic mechanism and (f) surface reconstruction of S-TiO₂. (g) FTIR spectra of PSTM before and after UV irradiation. (h) Schematic diagram of the hydrophilic surface cooling mechanism. (i–k) Surface temperature change curves of hydrophobic PSTM and hydrophilic PSTM under 0.5 sun, 1 sun, and 1.5 sun over 600s.

Here, the mechanism of the hydrophilic-hydrophobic conversion can be reasonably inferred, and it is largely attributed to TiO₂. [Fig. 7e](#) illustrates the mechanism of TiO₂ photocatalysis. When the TiO₂ surface is exposed to UV light, the electrons (e⁻) in the valence band (VB) are excited to the conduction band (CB), leaving the holes (h⁺) in the VB [39]. These holes have a strong oxidation capacity, which enables them to oxidize the water adsorbed on the surface of titanium dioxide to form a highly active hydroxyl radical (·OH), while the electrons excited to the CB can interact with molecular oxygen (O₂) adsorbed on the TiO₂ surface to form superoxide radical anions (·O²⁻), and then ·O²⁻ can further react with water to form ·OH [[40], [41], [42], [43], [44], [45], [46]]. These free radicals can further contribute to the degradation of the modified organic molecules, which may result in the exposure of the pure TiO₂ surface, which exhibits hydrophilic properties. At the same time, the holes can also react with the lattice oxygen of TiO₂ to form surface oxygen vacancies, which can lead to the adsorption of hydroxyl radicals and ultimately enhance the surface hydrophilicity [41,43,46]. When UV irradiation stops, the generation of photo-generated carriers ceases, and the existing electron-hole pairs subsequently recombine. This recombination process is accompanied by surface chemical restructuring ([Fig. 7f](#)), leading to the disappearance of surface oxidative active species (h⁺, ·OH, ·O²⁻) and the repair of surface oxygen vacancies by O₂ or H₂O from the environment. These changes collectively reduce the proportion of surface polar groups (—OH) and

increase the proportion of non-polar lattice structures (Ti—O), ultimately resulting in a decrease in surface energy. As shown in Fig. 7b, the surface energy significantly decreases from a high-energy state (42.94 mN/m) to a low-energy state (9.86 mN/m), weakening the hydrogen bonding between water molecules and the surface, thereby increasing the CA and ultimately restoring superhydrophobicity. Fig. 7g demonstrates the FTIR spectra of the PSTM surface before and after UV irradiation, which further confirms that UV can induce the transition from hydrophobic to hydrophilic on the PSTM. The peak at 3439cm^{-1} before UV irradiation is mainly the stretching vibration of —OH; however, this stretching vibration peak disappears after UV irradiation, indicating that UV induces the photogenerated holes to react with the —OH and H_2O on the surface to form $\cdot\text{OH}$, which leads to the depletion of —OH by the reaction. The absorption peaks at 2919cm^{-1} and 2851cm^{-1} before UV irradiation are mainly the C—H bonds in $-\text{CH}_3$ and $-\text{CH}_2$ of stearic acid, while the peak at 1631cm^{-1} is the C=O bond in $-\text{COOH}$, which indicates that the UV-induced photocatalytic reaction may decompose the organic matter on the surface, thus leading to the disappearance of the C—H and C=O peaks.

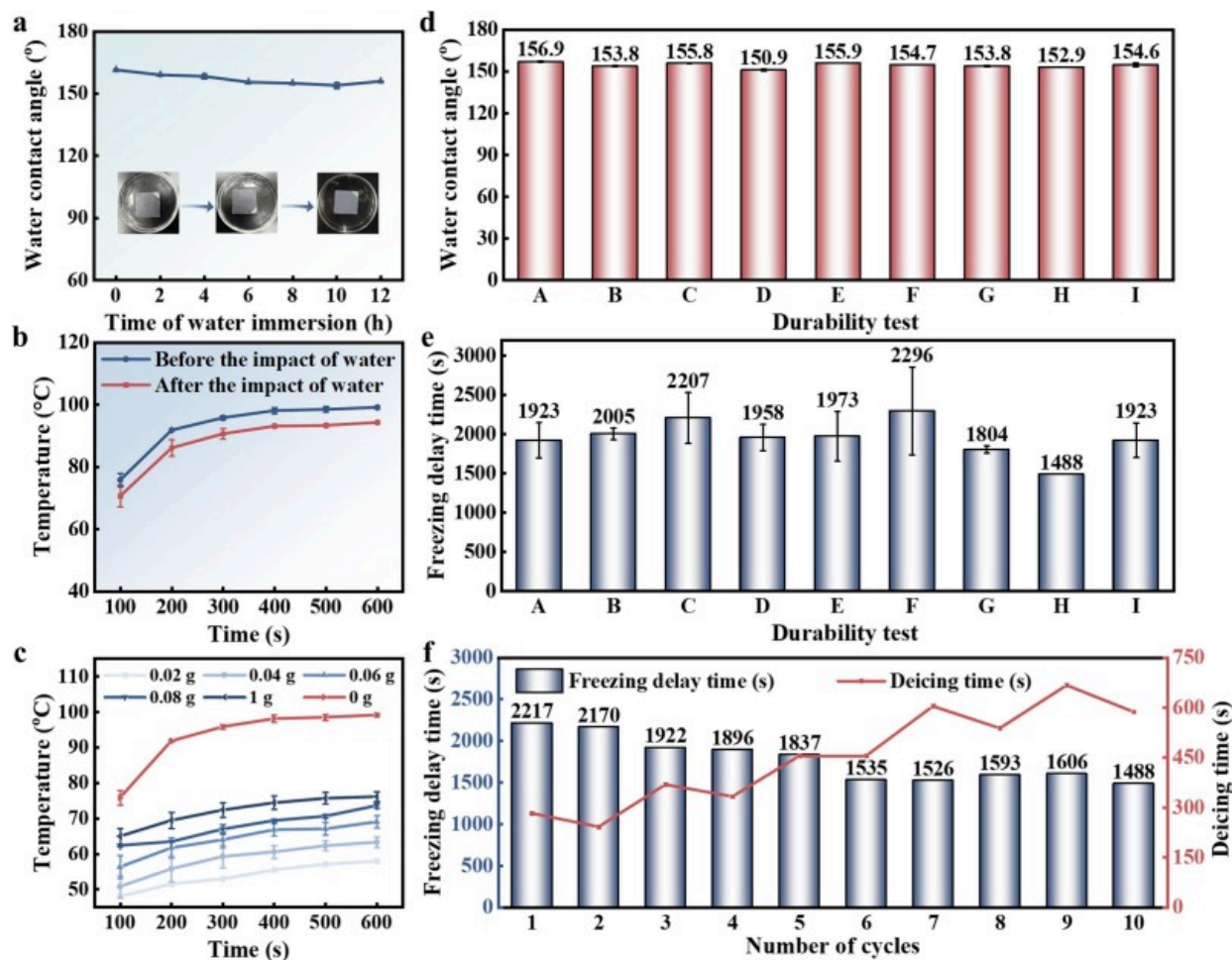
As shown in Fig. 7h, hydrophilic coatings form a water film (composed of liquid water molecules) on the surface by adsorbing moisture in the air or artificially added, and they take away most of the heat from the surface during the evaporation of the liquid water molecules to realize surface cooling, which is the principle of the evaporative cooling effect. The heat transfer efficiency between the coating and liquid droplets directly affects the cooling effect of the surface, and according to Fourier's law, the heat is always from the high-temperature region to the low-temperature region. A hydrophilic surface in close contact with the water film over a large area can accelerate the heat transfer from the coating to the droplets and thus accelerate the evaporation of water molecules, while a hydrophobic surface with a smaller contact area between the droplets and a higher thermal resistance leads to a limited heat transfer and thus slower evaporation of water molecules. In addition, hydrophilic surfaces can maintain the evaporation process by continuously absorbing water vapor in the environment, while hydrophobic surfaces repel water, droplets are easy to fall off, and the evaporation process is short-lived. Taken

together, the results of the above analysis indicate that hydrophilic surfaces are more effective than hydrophobic surfaces in cooling, and are expected to be used to solve the problem of surface overheating caused by high temperatures in summer. Therefore, we further evaluated the heat dissipation performance of hydrophilic surfaces covered with a water film. Figs. 7i-k, S18, and S19 show that under 0.5 sun, 1 sun, and 1.5 sun irradiation, the temperature of the hydrophilic surface was always lower than that of the hydrophobic surface. Especially after 600s of continuous 1.5 sun light irradiation, the temperature of the hydrophilic surface remained below 60°C, while the temperature of the hydrophobic PSTM exceeded 100°C, with a temperature difference of up to 53.8°C. This clearly demonstrates that hydrophilic surfaces not only exhibit excellent cooling performance, effectively preventing surface overheating, but their cooling effect becomes even more pronounced as light intensity increases.

3.5. Durability evaluation

Functional coatings are inevitably exposed to complex environments (such as particle erosion and rain impact) in practical applications, so the environmental durability of superhydrophobic materials is of critical importance. To assess the durability of PSTM, we conducted multiple tests, including sandpaper abrasion (Fig. S20), fine sand impact (Figs. S21 and S22), self-cleaning (Fig. S23), water impact (Figs. S24 and S25), water immersion, salt spray corrosion (Fig. S26), UV aging, cyclic anti-icing/deicing, and low-temperature high-humidity tests. Surprisingly, after undergoing the aforementioned tests, the CA of PSTM remained above 150° (Fig. 8d), demonstrating excellent hydrophobic stability and environmental adaptability; the CA after friction testing (156.9°) was slightly higher, primarily due to the synergistic effect of newly formed micro-roughness structures and low-surface-energy groups reinforcing the superhydrophobicity. In water immersion experiments, superhydrophobicity caused the silver mirror effect (Fig. S27), and as immersion time increased, the CA of PSTM fluctuated minimally (Fig. 8a); however, after water impact, the CA significantly decreased to 150.9°, indicating adverse effects on the multi-level rough microstructure. Nevertheless, due to the excellent light absorption capability of MWCNTs, the photothermal performance of PSTM only slightly decreased by

approximately 5 °C (Figs. 8b and S28). As is well known, contamination coverage severely weakens the photothermal performance of PSTM (Fig. 8c), disrupting the “light trap effect” and obstructing MWCNT photon absorption, thereby highlighting the importance of the self-cleaning capability conferred by superhydrophobicity (Fig. S29).



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Fig. 8. Durability evaluation of PSTM. (a) CA change curves of PSTM in the immersion test. (b) Surface temperature change curves of the PSTM under 1 sun before and after the water impact test. (c) The effect of dirt mass on the photothermal performance of PSTM

surfaces. (d) CAs and (e) freezing delay times of PSTM surface after each durability test, where A-I in the Figs represent sandpaper abrasion, fine sand impact, self-cleaning, water impact, water immersion, salt spray corrosion, UV aging, cyclic anti-icing/deicing, and low-temperature high-humidity test. (f) Delayed icing performance and deicing performance of PSTM in 10cycles of anti-icing/deicing tests.

Further testing showed that PSTM maintained excellent anti-icing performance after all durability tests (Fig. 8e), with delayed icing times exceeding 24min and reaching as high as 2296s after salt spray corrosion. This may be attributed to the synergistic effects of its superhydrophobic structure and residual chemical components. In contrast, the effects of 10cycles of anti-icing/deicing tests were more pronounced (Fig. 8f): as the number of cycles increased, both anti-icing and deicing performance declined, with the anti-icing time dropping to 1488s and the deicing time increasing to 587s by the tenth cycle. However, it still achieved effective delay of icing and deicing. Overall, PSTM maintains its critical hydrophobicity and anti-icing/deicing capabilities even after undergoing various harsh environmental tests, fully demonstrating its exceptional versatility and environmental durability in complex environments, providing strong support for its potential in practical anti-icing/deicing applications.

4. Conclusion

In summary, a new multifunctional anti-icing/deicing coating with superhydrophobicity, photothermal properties, and wettability conversion can be successfully prepared by a simple, low-cost and easy-to-control spraying method and can be used for practical anti-icing/deicing. In this design, the layered concave structure on the PSTM surface gives the coating excellent superhydrophobicity and light-capturing properties. The superhydrophobicity allows the PSTM to exhibit excellent anti-icing and anti-frosting properties. At -10°C , the freezing delay time of the PSTM surface is up to 2248s, which is about 19 times longer than that of a bare aluminum sheet. At -15°C , the droplet still shows bounce properties on the surface of the coating, which is expected to be used for low-temperature dynamic anti-icing. The outstanding photothermal properties enable

the surface ice to melt rapidly in only 283 s under 1 sun. In addition, the wettability conversion property of PSTM offers an innovative approach to solving the problem of overheating coated surfaces in the hot summer, which is expected to realize the multiseasonal application of anti-icing/deicing materials and broaden the application market of anti-icing/deicing materials.

CRediT authorship contribution statement

Fang Yao: Writing – review & editing, Validation, Supervision, Data curation. **Jue Wei:** Writing – review & editing, Methodology, Formal analysis. **Yujie Xu:** Writing – review & editing, Data curation. **Hao Tu:** Writing – review & editing, Data curation. **Min Huang:** Writing – review & editing, Validation. **Qifeng Tang:** Writing – review & editing, Validation. **Shuai Zhao:** Writing – review & editing, Data curation. **Jian Wang:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Sichuan Science and Technology Program (No. [2024NSFSC0255](#)).

Appendix A. Supplementary data

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Supplementary Video S1. Video. S1. Dynamic anti-icing process of PSTM at -10°C (30 min).

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Supplementary material 1. Chemicals and materials; Micro-morphology and elemental analysis; Wetting and surface energy measurement; Anti-icing test; Anti-frosting test; Low-temperature resistance test; Ice adhesion strength test; Photothermal performance test; Photothermal deicing test; Switchable wettability test; Friction test; Falling sand test; Self-cleaning test; Water impact test; Water immersion test; Salt spray corrosion test; UV aging test; Thermal cycling test; High humidity test. Supplementary results and discussion.

Data availability

No data was used for the research described in the article.

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